Literature Answer to Problem 1.13 of arXiv:math/0508650

FA/Banach run packet

2026-06-18

Status

This is a literature-already-answered packet. It records that Problem 1.13 from Dilworth–Odell–Sari [1] was answered in the later paper of Dodos [2]. It is not an original proof from this run.

Original Problem

Problem 1.13 appears in Section 1 of [1]. In the notation of that paper, it asks:

> If $SP_w(X)$ is uncountable, must there exist $\{(\tilde{x}_i^\alpha)_{i=1}^\infty : \alpha < \omega_1\} \subseteq SP_w(X)$ which is either strictly increasing with respect to α, strictly decreasing, or consists of mutually incomparable elements?

Supporting Answer

Dodos [2] explicitly introduces the same question as the problem posed by the authors of [1], citing it as [1, Problem 1.13]. The abstract states that, for every separable Banach space X, either $SP_w(X)$ is countable or $SP_w(X)$ contains an antichain of cardinality continuum, and says that this answers a question of Dilworth, Odell, and Sari.

Literature Answer. *Problem 1.13 of [1] has a positive answer for separable Banach spaces. If $SP_w(X)$ is uncountable, then $SP_w(X)$ contains a family of cardinality continuum whose elements are pairwise incomparable in the domination order.*

Literature identification. Theorem 1 of [2] states that, whenever $SP_w(X)$ is uncountable, there is a Cantor-tree family $(x_t)_{t\in 2^{<\mathbb{N}}}$ in X such that different branches generate spreading models incomparable under domination. Corollary 1 of [2] records the direct consequence: $SP_w(X)$ contains an antichain of size continuum.

Since continuum has cardinality at least ω_1, this continuum-sized antichain contains an ω_1-indexed subfamily. That subfamily satisfies the third alternative in Problem 1.13 of [1], namely mutual incomparability. Thus the later paper answers the original problem. $\square$

Scope Limitations

This packet answers only Problem 1.13 of [1]. It does not address the countable-spreading-model Problem 1.6, nor the broader set-theoretic semilattice Problem 1.14 in the same source paper.

Verification Notes

The source and supporting papers are distinct. The supporting paper explicitly cites [1, Problem 1.13], so this belongs in `literature_already_answered` rather than `literature_implied_answers`. The local duplicate search covered arXiv id `0508650`, title keywords `Lattice structures and spreading models`, and phrases `Problem 1.13`, `uncountable SP_w`, and `antichain of spreading models`.

References

[1] S. J. Dilworth, E. Odell, and B. Sari, *Lattice structures and spreading models*, arXiv:math/0508650.

[2] P. Dodos, *On antichains of spreading models of Banach spaces*, arXiv:0805.2038.